Simple Linear Regression

Can correlation predict exact sales?
Not by itself. Regression provides an equation that can be used to estimate sales from advertising spend.

What is Simple Linear regression?
It estimates the relationship between one independent variable and another dependent variable.
Technically:
It models the conditional mean of a dependent variable Y as a linear function of one predictor X

Example:
A company wants to understand how adverstising expenditure relates to sales.
Indpendent Varaible(X) : Advertising spend
Dependent Variable(Y) : Sales

If we spend ₹60,000 on advertising, what sales value we might expect?

Correlaion vs regression

Correlation	**Regression**
Measures strength and direction	Models a relationship
Produce vlue between -1 to +1	Produces an equation
Does not directly predict Y	It can estimate Y from X
Treats the variables symmetrically	Assigns predictor and response rols.

Regression does not automatically prove casusation. Predictions are reliable only to the extent that the model and its assumptions are appropriate.

Regression Equation

$Y = aX+b$ $Y = a+bX$

Y = Predicted value of Y
X = Independent variable
a = Intercept
b = Slope

Example Correl = 0.996623

Month	Spend(X)	Sales(Y)	X^2	XY
1	10	20	100	200
2	20	28	400	560
3	30	42	900	1260
4	40	50	1600	2000
5	50	65	2500	3250
6	60	72	3600	4320
7	70	80	4900	5600
8	80	90	6400	7200

360 447 20400 24390

Calcualting Slope and Intercept

Step 1: Calculate the slope

$$b = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

n = 8, ∑X=360,∑Y=447
∑X^2=20400,∑XY = 24390

Using compuational formula:

$$b = \frac{n\sum XY - (\sum X)(\sum Y)}{n\sum X^2 - (\sum X)^2}$$

$$b = \frac{8*24390 - 360*447}{8*20400 - 360*360}$$

195120 160920
163200 129600
34200 33600

$$b = \frac{195120-160920}{163200-129600}$$

1.017857

$$b = \frac{34200}{33600}$$

b = 1.017857

Advertising spend is measured in ₹ thousands, and sales are also measured in ₹ thousands
A slope 1.017857 means:
For each additional ₹1,000 in the advertising expenditure, the model predicts aproximately ₹1018 in additional sales, on average, within the model relationship

Calculate the intercept

Formula:

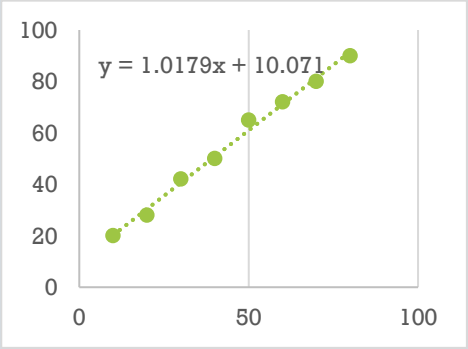
a = Ybar-bXbar

Xbar = 360/8 = 45

Ybar = 447/8 = 55.875

a = 55.875 - 1.017857*45

a = 10.07144



Write Regression Equation

Y = a +bX

Y = 10.071 + 1.0178X

Prediction Using Regression

Y = 10.071 + 1.0178X

Example: Advertising Spend = ₹60,000

Substitute:

Y = 10.071 + 1.0178 *60 61.068
Y = 10.071 + 61.068
Y = 71.139

Predicted sales are approximately = ₹71,139

Example: Advertising Spend = 90,000
Y = 10.071 +1.0178*90
101.673
Predicted sales are approximately ₹101,673

Limitations

The observed advertising expenditure ranges from ₹10,000 to ₹80,000.
Predicting sales at ₹90,000 is extrapolation because it is outside the observed range.

The model's linear pattern may not continue at higher spending levels. The company should treat this estimation catiously.

Practical Business Uses:

1. Forecast sales for a proposed advertising budget.
2. Estimate demand based on price.
3. Estimate production output based on machine operating hours
4. Estimate customer spending based on visits.

R Squared

It tells us what proprtion of the variation in the dependent variable is explained by the regression model.

Techincally

It is the coeficient of determination, calculated as the proprtion of total variation in the observed response explained by the fitted regression model.

Formula:

$$R^2 = 1 - \frac{SS_{Residual}}{SS_{Total}}$$

Where:

$SS_{Residual}$ Sum of Squared differences between actual and predicted values.
SS_{Total} Sum of Squared differences between actual values and their mean.

Business Example:

Spend(X)	Sales(Y)	Predict	(Y-PredictY)^2
10	20	20.249	0.062001
20	28	30.427	5.890329
30	42	40.605	1.946025
40	50	50.783	0.613089
50	65	60.961	16.31352
60	72	71.139	0.741321
70	80	81.317	1.734489
80	90	91.495	2.235025

0.991581 29.5358 0 29.5358

A company wants to understand the relationship between advertising spend and sales
R^2 = 0.85
The fitted model explains 85% of the observed variation in sales in the dataset
It may reflect other factors, random variations, measurment error, or a limitation by the model.

Suppose

$SS_{Residual}$ 1000

SS_{Total} 150

R^2 = 1- (150/1000)
R^2 = 1- (150/1000)
R^2 = 1-0.15 = 0.85
85%

Adjusted R-- Meaning and calculation